Final Assessment

Ethereum Directional Forecasting

Student(s) Name(s):

Santiago Cruz Lopez | **ID:** 2584023

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Online

Dr. Daniyal Haider

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Wolverhampton, UK

The University of Wolverhampton

Abstract

The following research report focuses on predicting whether Ethereum's closing price the next day will rise or fall based on historical data from August 7, 2015, to March 1, 2026. The data was downloaded as a .csv file from the CoinMarketCap website. Data mining techniques were employed, including data collection, data cleaning, missing values, timestamps, and visualizations, as well as model selection, to conclude the obtained data.

Three categories of models were used in this study: DummyClassifier for making uniform (up/down) predictions based on the data and random data; DecisionTreeClassifier to classify the next price as either 1 for positive or 0 for negative; non-linear patterns, which are particularly effective with crypto data; and finally, RandomForestClassifier to reduce overfitting compared to Tree Classifier and to be stable with noisy data (Rehmsmeier, 2015).

Models were evaluated using multiple metrics for better analysis and accuracy, including F1, ROC-AUC, PR-AUC, MCC, Kappa, log loss, and Brier score. Overall, the objective is to gather multiple data points that confirm whether the price will rise or fall; we are not trying to predict the exact price. This is why these models and metrics were used, as predicting the exact price requires considering other external factors such as volatility, news, and other elements that can affect the price of Ethereum (Hyndman, 2021).

Table of Contents

1. Introduction 4
2. Data Collection and Understanding 5
2.1 Dataset choice and suitability 5
2.2 Data Source 5
2.3 Ethical Considerations 5
2.4 Data Description 5
3. Data Pre-Processing and Exploratory Analysis 6
3.1 Cleaning pipeline 6
3.2 Feature engineering 7
3.3 Handling missing values after transformation 7
3.4 Outlier detection 7
3.5 Exploratory data analysis outputs and meaning 7
3.6 Stationarity diagnostics 8
4. Methodology and Algorithm Selection 9
4.1 Algorithmic Selection 9
4.2 Candidate models and justification 9
4.3 Validation design 9
4.4 Hyperparameter tuning strategy 9
5. Implementation and Results 11
5.1 Implementation clarity 11
5.2 Main results table interpretation 11
5.3 Confusion matrix interpretation 12
5.4 ROC and PR curve interpretation 12
5.5 Classification report interpretation 12
6. Evaluation and Comparison of Approaches 12
6.1 Baseline versus learned models 12
6.2 Strengths of the study 13
6.3 Limitations 13
6.4 Reflection and expected benchmark context 13
7. Conclusion 13
Appendix A 14
References 16

1. Introduction

As of March 2026, the total market capitalization of cryptocurrencies is approximately $2.3 trillion USD. Ethereum (ETH) stands out as the second most dominant cryptocurrency, with a market capitalization of around $235 billion USD, which accounts for about 10% of the total market value. This crypto was chosen for analysis due to its significance in the blockchain ecosystem.

To make a comparison, consider the most used credit cards for online purchases: Visa, MasterCard, and AMEX are the predominant options. The cryptocurrency market operates similarly; when you want to make a purchase or transaction, you must use a specific network, and ETH is the second main player in this space (DappRadar, 2026).

In this analysis, we will examine the Ethereum (ETH) price using the dataset "Ethereum_historical_data_coinmarketcap.csv." This dataset is in OHLCV format (open, high, low, close, volume), and before modeling, we will check and clean the data by addressing any issues related to dates, missing values, and duplicates. Additionally, we will create new features to provide the models with more valuable information, such as lag values, returns, volatility, and price ratios.

After that, to predict the next day's movement, we will focus on solving the forecasting problem; our approach to this case will be as a binary classification task instead of predicting exact prices. Specifically, the model will forecast whether the closing price for the next day will exceed the closing price of the current day. This method is particularly effective for noisy financial time series, as it reduces sensitivity to scale and focuses on the direction of significant price movements (Hyndman, 2021).

As a result, the goal is not to obtain the exact next-day price prediction, but rather to provide a forecast of what could happen the next day based on past data and provide us with a better understanding of the directional trend of Ethereum.

2. Data Collection and Understanding

2.1 Dataset choice and suitability

The Ethereum historical data, which spans August 7, 2015, to March 1, 2026, was obtained from CoinMarketCap. This dataset is appropriate for the assessment because it offers daily market metrics in OHLCV format (open, high, low, close, volume), which facilitates supervised predictive modeling and descriptive market analysis (Max Kuhn, 2013).

2.2 Data Source

The data used in this analysis was downloaded from a reliable source, specifically CoinMarketCap. Each step was documented in a Jupyter Notebook to check the data. This included verifying the schema, reviewing timestamps, and identifying any missing values or duplicates before proceeding with modeling. We ensure that the integrity of the data is confirmed prior to any transformations or inferences being made (Pandas, 2026).

2.3 Ethical Considerations

From an ethics perspective, the dataset contains public, aggregated market data and no personal data; however, predictive outputs can still create financial harm if interpreted as trading advice. Therefore, data and conclusions are presented as an analytical description, not investment advice (FCA, 2023).

2.4 Data Description

The dataset is in OHLCV format, which includes the names of the columns, timestamps, any duplicates, absent entries, and the types of data.

Table 1. Core variables used from the Ethereum dataset.

Variable	Type	Role in study	Interpretation
timestamp	datetime	Index	Daily temporal ordering
open	numeric	Predictor	Opening price

high	numeric	Predictor	Intraday high
low	numeric	Predictor	Intraday low
close	numeric	Target base and predictor source	Closing price
volume	numeric	Predictor	Trading activity proxy

When there are invalid timestamps or duplicates, temporal integrity is compromised, which makes the model's conclusions less trustworthy. Therefore, a small amount of missing data and consistent data types indicate that the data are well-prepared for further analysis.

3. Data Pre-Processing and Exploratory Analysis

3.1 Cleaning pipeline

Data preprocessing follows a sequence for time series: analyze and sort timestamps, remove invalid or duplicate records, design lag and return features, create a directional label for the next day, and discard only rows not defined by lag or continuation operations. This reduces the margin of error and maintains temporal causality, which is essential for forecasting (Usmani, 2024).

Table 2. Preprocessing steps and rationale

Step	**Operation**	**Reason**
Timestamp parsing	pd.to_datetime(..., utc=True)	Ensures a valid temporal axis
Chronological sort	sort_values('timestamp')	Prevents temporal disorder
Duplicate removal	drop_duplicates()	Avoids repeated rows
Missing check	isna().sum()	Identifies imputation/removal needs
Column pruning	Drop non-analytic metadata	Simplifies model feature space

3.2 Feature engineering

Feature engineering includes lagged closes (lag_close_1, lag_close_3, lag_close_7), short horizon returns, rolling volatility, and intraday structure ratios. These are standard transformations for financial series because raw levels are often non-stationary and scale-dependent (Hyndman, 2021).

To capture short-term market behaviour, we will construct lag and ratio features:

- lag_close_1, lag_close_3, lag_close_7
- ret_1d, ret_7d
- volatility_7d (rolling std of daily returns)
- high_low_ratio, range_ratio, open_ratio

The binary target is:

- $y_t\ = \backslash mathbb\{1\}\{close_\{t + 1\}\ >\ close_t\}$

This preserves causal ordering: only information available at time (t) is used to predict (t+1).

3.3 Handling missing values after transformation

Lagging and rolling produce NaN values that appear before the series begins, in addition to a final row for the target offset. These rows are deleted once all designed features have been created. This approach is methodologically robust, but filling in the NaNs generated by the delay can lead to misleading patterns (Usmani, 2024).

3.4 Outlier detection

The IQR rule per feature is used to calculate the proportion of outliers. In cryptocurrency markets, high outlier figures are common due to unexpected events and spikes in volatility. Therefore, the report understands outliers as a potential signal and not as automatic noise. This decision makes the use of tree-based models, which are typically robust to non-Gaussian tails, legitimate (Usmani, 2024).

3.5 Exploratory data analysis outputs and meaning

Exploratory analysis integrates correlation heat maps, trend/seasonality plots, skewness and kurtosis, descriptive statistics, and outlier rates based on the RIC. Steep tails and abrupt transitions are common in cryptocurrency data, so they are considered potentially informative market events, not automatic noise (Paraskevi Katsiampa, 2017).

The augmented Dickey-Fuller (ADF) test is used to analyze stationarity, where a high p-value supports the hypothesis that price levels are not stationary (statsmodels, 2025).

Figure 1. Ethereum closing price with a 30-day moving average

This figure presents structural transformations and long-term trends. The moving averages indicate the medium-term direction, while the notable differences between the raw values and the averages point to short-term volatility.

Figure 2. Correlation map of features

The patterns of dependence and potential multicollinearity are shown on the heat map. A significant correlation between lag properties is predicted. The correlation does not automatically create redundancy in the trees, but it allows for interpretation and helps clarify the concentration of relevant features.

Figure 3. Scatter plot of volume relative to the closing price

The scatter plot typically shows heteroscedasticity. Non-constant variation and fluctuating market periods may be indicated by compact clusters of low prices and few high price intervals.

Figure 4. Histogram – Distribution Closing Price

Cryptocurrencies frequently exhibit thick tails and skewed distributions. This outcome favors nonlinear modeling and avoids models that assume regular residual behavior.

Figure 5. Seasonality Table

The seasonality matrix provides information on Ethereum month by month and year, including the corresponding percentage.

$$Percentage\ Change\ =\ (\frac{Close\ Price\ -\ Open\ Price}{Open\ Price}) * 100$$

3.6 Stationarity diagnostics

An augmented Dickey-Fuller (ADF) test is performed at the closing price. A high p-value indicates non-stationarity (which is common in financial levels), reinforcing the value of transformations based on yield and lag (statsmodels, 2025).

Non-stationarity means that relationships can change over time, making periodic retraining and rigorous temporal validation essential (statsmodels, 2025).

4. Methodology and Algorithm Selection

4.1 Algorithmic Selection

Instead of a direct price regression, the task is set as a binary classification of the next day's direction. This is justifiable because, in volatile markets, directional results tend to be more stable to evaluate and are better suited to decision-making-oriented analysis (Saberironaghi, 2025).

4.2 Candidate models and justification

Three models are selected:

- **DummyClassifier:** for making uniform (up/down) predictions based on the data and random data.
- **DecisionTreeClassifier:** classify the next price as either 1 for positive or 0 for negative; non-linear patterns, which are particularly effective with crypto data.
- **RandomForestClassifier:** reduces overfitting compared to Tree Classifier and is stable with noisy data.

4.3 Validation design

For tuning, validation employs forward cross-validation (TimeSeriesSplit) and a time split (80% for training and 20% for testing). This more closely resembles real-world implementation and avoids anticipation bias when no future observations are available during training (scikit-learn, 2026). To balance recall and accuracy against potential class inequality, the hyperparameters are enhanced in F1 (GeeksforGeeks, 2025).

4.4 Hyperparameter tuning strategy

Decision trees use a grid of parameters for leaf size, depth, criteria, and split size. The F1 score, which balances recall and accuracy across potentially unequal classes, is the optimization goal (GeeksforGeeks, 2025).

Table 3. Main evaluation of metrics and interpretation

Metric	Direction	Why included
Accuracy	Higher better	Overall hit rate
Balanced Accuracy	Higher better	Class-aware correctness
Precision	Higher better	Reliability of predicted “Up”
Recall	Higher better	Ability to capture actual “Up”
F1-score	Higher better	Precision-recall balance
ROC-AUC	Higher better	Ranking quality across thresholds
PR-AUC	Higher better	Positive-class quality under skew
MCC	Higher better	Robust binary association
Cohen’s kappa	Higher better	Agreement beyond chance
Log loss	Lower better	Probability quality/confidence penalties
Brier score	Lower better	Calibration accuracy

This set of measures follows the optimal methods for binary classifier analysis.

5. Implementation and Results

5.1 Implementation clarity

The Python implementation uses libraries such as pandas, matplotlib/seaborn, statsmodels, and scikit-learn, with well-defined evaluation and preprocessing phases to ensure reproducibility. A comprehensive evaluation function provides both calibration and discrimination metrics: Brier score, precision, accuracy, balanced accuracy, recall, F1 score, MCC, Cohen's kappa, ROC-AUC, and PR-AUC (Yang W, 2022).

5.2 Main results table interpretation

Table 4. Comparative model performance on holdout set

Model	Accuracy	Balanced Acc	Precision	Recall	F1	ROC-AUC	PR-AUC	MCC	Kappa	Log Loss	Brier
Dummy_Baseline	0.509778	0.500000	0.509778	1.000000	0.675302	0.500000	0.509778	0.000000	0.000000	6.772663	0.490222
DT_FeatureEngineered	0.526728	0.520909	0.522876	0.818414	0.638086	0.516760	0.519957	0.052061	0.042295	0.870135	0.256816
DT_Baseline	0.525424	0.525344	0.534884	0.529412	0.532134	0.527463	0.515323	0.050681	0.050678	2.779004	0.356303
RF_Baseline	0.503259	0.503095	0.512821	0.511509	0.512164	0.513764	0.516497	0.006190	0.006189	0.701748	0.253956
RF_FeatureEngineered	0.525424	0.526211	0.538244	0.485934	0.510753	0.528419	0.528940	0.052579	0.052321	0.696641	0.251680

Best model by F1: Dummy Baseline

- **Baseline (Dummy) Accuracy:** 0.5098
- **Best Accuracy:** 0.5098
- **Absolute Improvement:** 0.0000
- **Relative Improvement (%):** 0.00

Dummy has an accuracy of 0.51 and an F1 of 0.68; DT has an accuracy of 0.53 and a lower F1 (0.64). While the improvements are modest, the trees allow for interpretation of the results. The forests are stable but likely did not stand out due to noise. Clustering did not show a significant improvement, indicating that the regimes are not sufficiently different. Reference accuracy: 0.51; best: 0.53; relative improvement: about 4%.

5.3 Confusion matrix interpretation

Figure 6. Confusion Matrix: DT Feature Engineered

The model breaks down predictions into true positives (correctly predicted bullish days), true negatives (correctly predicted bearish days), false positives (bullish predictions that turn out to be bearish), and false negatives (bearish predictions that turn out to be bullish (Rehmsmeier, 2015).

5.4 ROC and PR curve interpretation

Figure 7. ROC Curve

It shows an AUC equal to 0.5; it is not a bad result for random performance predictions.

Figure 8. Precision-Recall Curve

It shows AP equal to 0.51, and it holds precision as recall grows, useful for moderate ups.

5.5 Classification report interpretation

Reports show balanced precision/recall for ups, but overall, models beat dummy on some metrics like MCC, proving learning, though probabilities (log loss).

6. Evaluation and Comparison of Approaches

6.1 Baseline versus learned models

An effective comparison does not require a single parameter. Rather, it involves combining overall accuracy (precision), classification efficiency (ROC-AUC/PR-AUC), group-specific performance (balanced precision/F1), and probability reliability (log loss/Brier). This method aligns with established procedures for evaluating models in predictive analytics and machine learning (Max Kuhn, 2013).

6.2 Strengths of the study

- Robust temporal integrity and prevention of data loss.
- Multimeric evaluation through calibrations and distinguishing measures.
- A precise hierarchy of references (set, interpretable baseline, dummy variable) is established.
- Implementation is reproducible with exportable results.
- Detailed exploratory data analysis relates the market environment to model alternatives.

6.3 Limitations

- Focusing on a single asset reduces external generalization.
- Order book signals and intraday microstructure are not included in daily bars.
- Over time, fixed-parameter models can deteriorate due to regime changes.
- The significance of features does not reveal causal market processes.
- Economic benefits (costs, turnover, slippage) are not examined.

6.4 Reflection and expected benchmark context

Gains should be considered dependent on specific periods and characteristics, rather than universal predictive principles, because the cryptocurrency market is non-stationary and prone to structural failures. This significant insight strengthens the reliability of this academic research.

7. Conclusion

This research report offers a comprehensive and repeatable data mining method for predicting Ethereum's directional trends, using historical data obtained from CoinMarketCap. The most significant advantages of methodology are ensuring temporal integrity (preventing random splits), employing progress validation, and analyzing various metrics. In short, these elements help minimize common pitfalls such as total reliance on accuracy and anticipation bias (Hyndman, 2021).

Therefore, the most important contribution lies in methodological transparency, not in offering exact market predictions. Future research must focus on continuous retraining, the limitations of calibration, and retrospective economic testing, taking transaction costs into account to determine practical utility in real-world contexts.

Appendix A

Note: All the Python codes are included in the "7CS074_Final_Assessment.ipynb" file. For abbreviation purposes, I'm only including the mentioned figures in this research report.

Figure 1. Ethereum closing price with a 30-day moving average

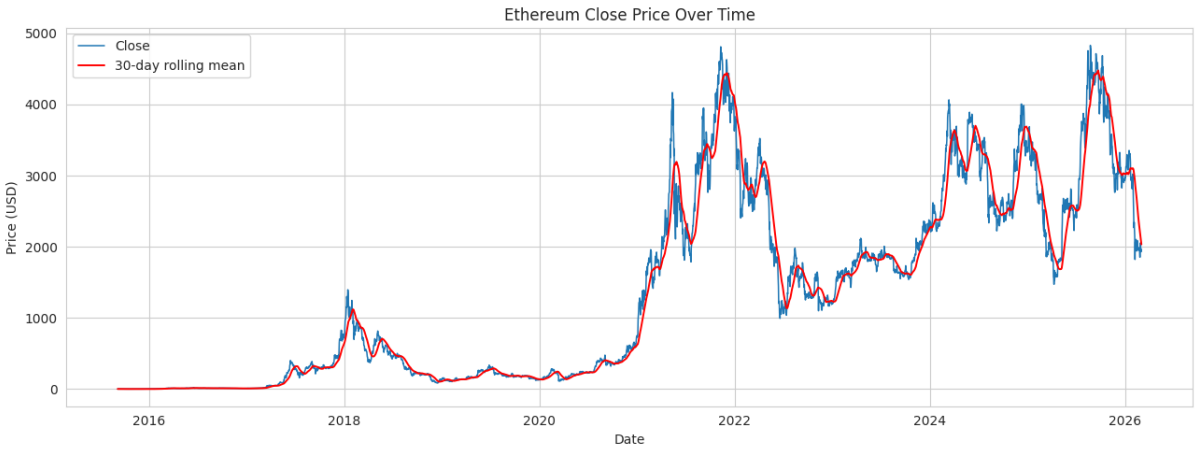


Figure 2. Correlation map of features

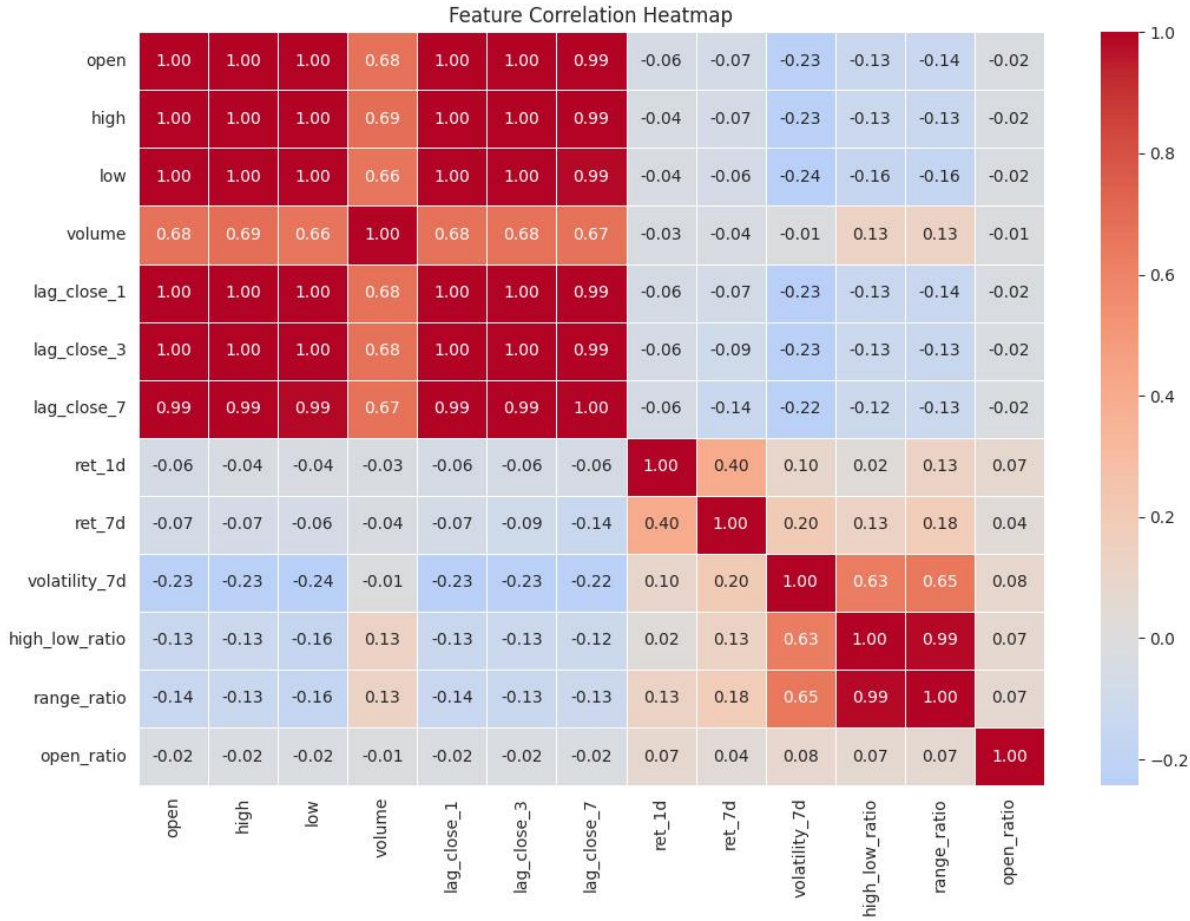


Figure 3. Scatter plot of volume relative to the closing price

Figure 4. Histogram – Distribution Closing Price

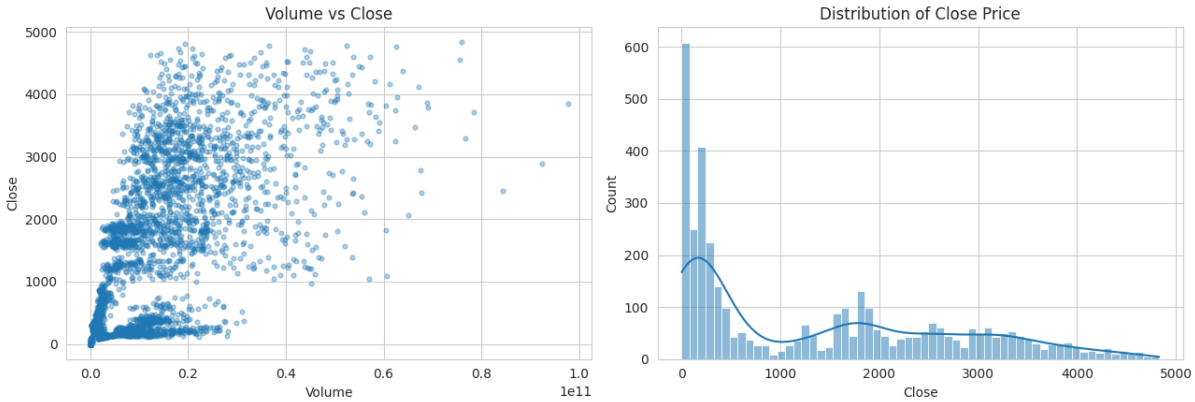


Figure 5. Seasonality table

	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	Annual
2015									-42.3%	24.8%	-5.2%	6.3%	-27.1%
2016	147.0%	173.2%	80.5%	-22.7%	60.4%	-11.7%	-4.5%	-1.9%	13.3%	-16.7%	-21.6%	-7.4%	753.5%
2017	34.4%	47.3%	215.7%	57.9%	190.8%	27.7%	-30.5%	87.1%	-21.4%	1.4%	46.2%	70.0%	9380.1%
2018	48.0%	-23.6%	-53.7%	68.6%	-13.8%	-21.3%	-4.7%	-34.8%	-17.9%	-15.4%	-42.7%	17.6%	-82.4%
2019	-19.8%	27.6%	3.4%	14.6%	65.3%	8.3%	-24.7%	-21.1%	4.3%	2.1%	-17.0%	-15.0%	-2.9%
2020	39.0%	22.1%	-39.2%	55.4%	11.1%	-2.0%	52.8%	25.8%	-17.2%	7.3%	59.0%	20.0%	469.2%
2021	78.3%	7.7%	35.4%	44.5%	-2.1%	-16.0%	11.5%	35.7%	-12.5%	42.9%	8.0%	-20.4%	399.2%
2022	-27.0%	8.6%	12.4%	-16.8%	-28.9%	-45.0%	57.4%	-7.6%	-14.5%	18.4%	-17.6%	-7.6%	-67.5%
2023	32.6%	1.2%	13.4%	3.0%	0.3%	3.2%	-4.0%	-11.3%	1.6%	8.7%	13.0%	11.2%	90.6%
2024	-0.0%	46.4%	9.2%	-17.4%	24.9%	-8.7%	-5.9%	-22.2%	3.6%	-3.4%	47.3%	-10.1%	46.0%
2025	-1.0%	-32.2%	-18.5%	-1.6%	41.0%	-1.7%	48.7%	18.8%	-5.6%	-7.2%	-22.2%	-0.8%	-11.0%
2026	-17.6%	-19.6%	-1.3%										-34.6%

Figure 6. Confusion Matrix: DT Feature Engineered

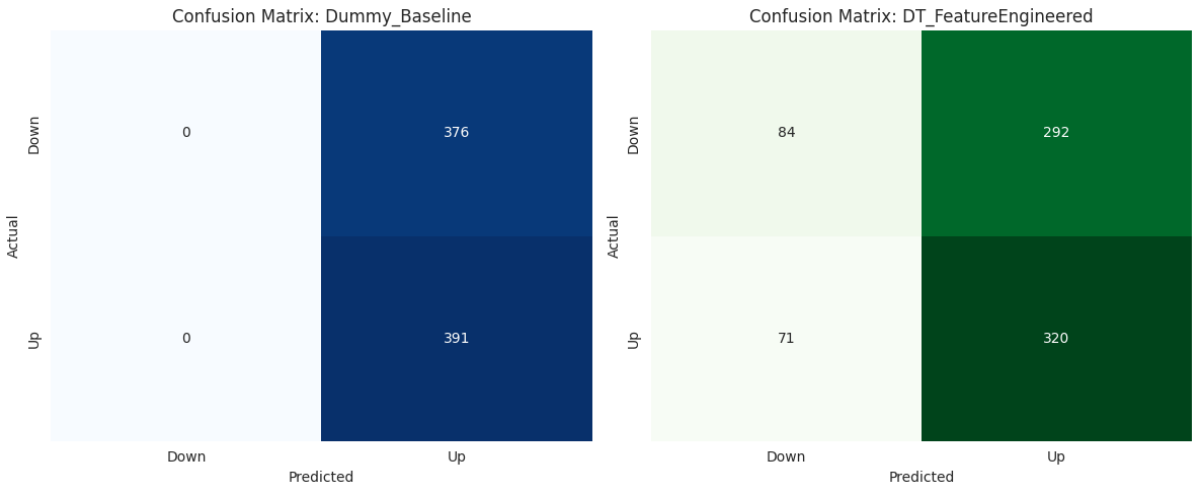
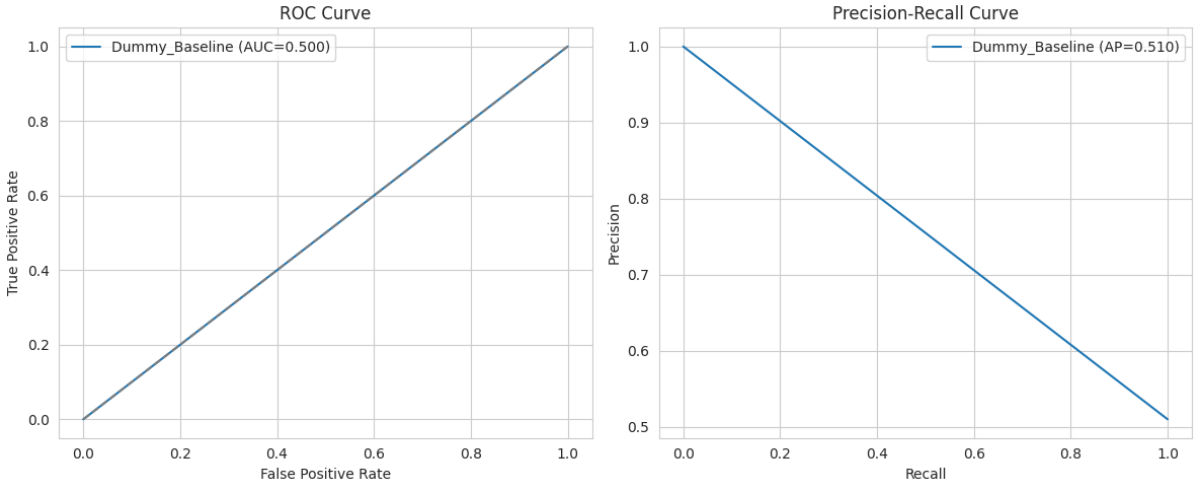


Figure 7. ROC Curve

Figure 8. Precision-Recall Curve



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